

Toolkit for adaptation of the WHO recommendations for a positive pregnancy and postnatal experience



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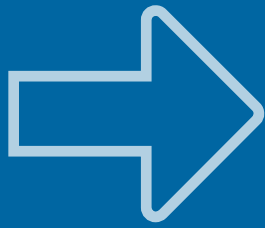
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Acronyms and abbreviations

ANC	antenatal care	LLINs	long-lasting insecticidal nets
ASB	asymptomatic bacteriuria	MUAC	mid-upper arm circumference
BAT	baseline assessment tool	NGO	non-governmental association
CHW	community health worker	PNC	postnatal care
CMIS	community management information system	PrEP	pre-exposure prophylaxis (for HIV)
DHS	Demographic and Health Surveys	QES	qualitative evidence synthesis
FHR	fetal heart rate	SIDS	sudden infant death syndrome
HIV	human immunodeficiency virus	STI	sexually transmitted infection
HMIS	health management information system	SWOT	strengths, weaknesses, opportunities, threats
HRP	UNDP/UNFPA/UNICEF/WHO/World Bank Special Programme of Research, Development and Research Training in Human Reproduction	TB	tuberculosis
IFA	iron and folic acid	TT/TD	tetanus toxoid/tetanus diphtheria vaccine
ILO	International Labour Organization	USAID	United States Agency for International Development
IPT	intermittent preventive treatment	VMMC	voluntary medical male circumcision
IPV	intimate partner violence	WHO	World Health Organization



Introduction

About the toolkit

The World Health Organization (WHO) has published recommendations on antenatal care for a positive pregnancy, intrapartum care for a positive childbirth experience, and maternal and newborn care for a positive postnatal experience (1–3). Together, these recommendations provide integrated guidance on the provision of high-quality essential care to women and newborns across the pregnancy to postnatal continuum.

The recommendations on routine antenatal care (ANC) for all pregnant women and adolescent girls were first released in 2016 (1) and covered maternal and fetal assessment, preventive health measures, nutrition, and interventions for common physiological symptoms during pregnancy, as well as health systems interventions aimed at increasing ANC service use and improving the quality of care on offer.

In 2022, WHO launched its recommendations on maternal and newborn care for a positive postnatal experience (3), which updates and expands on previous recommendations (4). These postnatal care (PNC) recommendations relate to care beginning after the birth of the baby and continuing up to six weeks (42 days)

after birth. They cover maternal care, such as maternal assessment, interventions for common physiological signs and symptoms, preventive health measures and interventions to address mental health, nutrition and contraception in the first weeks following childbirth. The PNC recommendations also cover newborn care, including newborn assessment, preventive health measures, infant growth and development, nutrition and breastfeeding. Recommendations on health systems and health promotion interventions to improve the uptake and quality of PNC are also included. The updated PNC recommendations are intended to complement the existing WHO recommendations on the care of the woman and the newborn after birth.

The 2016 ANC recommendations set out an eight-contact ANC model, while the 2022 PNC recommendations propose a four-contact PNC model. The models place the woman and the baby at the centre of care and highlight the importance of essential care practices (including self-care and family care practices), continuing support from health services, and the support of partners, caregivers and family. Both models seek to prioritize health and well-being using person-centred and human rights approaches and respond to the complex nature of the issues associated

with the practice, organization and delivery of care within a variety of health systems and settings.

Introducing the WHO ANC and PNC models in different countries requires adapting them to existing health systems (5). Implementing these person-centred models may require extensive reform of existing health programmes, given the difficulties of integrating health programmes into diverse health systems, the complexities of adapting recommendations to particular contexts, and the known barriers to applying guidelines (6). Additionally, the renewed focus on the care of women and newborns in the first 24 hours after birth set out in the PNC model, as well as the greater attention to newborn screening, maternal mental health and discharge preparedness, will require thoughtful consideration of barriers to implementation at policy, health systems and user levels.

This toolkit will encourage country teams to systematically (i) review the ANC and PNC recommendations and adapt them to their context and (ii) update their ANC and PNC policies according to the WHO guidelines. The aim of the toolkit is to facilitate the building of country-specific packages of ANC and PNC interventions that are in accordance with WHO evidence-based recommendations and tailored to the individual contexts. The toolkit is also designed to highlight country-specific factors that are likely to influence (positively or negatively) the implementation of a country's package of ANC and PNC interventions. Such factors should be considered when planning for implementation. In addition, the inclusion in the toolkit of an exercise for mapping international human rights policy instruments against national legislation encourages country teams to consider the broader enabling environment for implementing the recommendations. The toolkit was envisioned for use at the national level. However, with slight modifications, it can also be applied at subnational levels. For example, the

situational analysis of the current provision of ANC and/or PNC could be conducted at district level, for planning purposes and to aid the implementation of the national package.

How the toolkit was developed

A first ANC recommendations adaptation toolkit was developed with the aim of helping national governments to adapt the 2016 WHO recommendations to their context (6). It used an iterative process including development of a draft toolkit, user testing in four countries (Burkina Faso, India [two states: Assam and Tamil Nadu], Rwanda and Zambia), and refinements based on feedback from country users.

WHO has revised and updated these tools originally designed to support the adaptation of the ANC recommendations so that they include updates to the ANC recommendations (7–10), as well as support the adaptation of the 2022 PNC recommendations. The present version of the toolkit was tested in Guinea and Rwanda, during national adaptation processes of the WHO recommendations, to assess its usefulness, adherence to the proposed aim and any barriers to its use.

Who is the toolkit for?

The primary target audience for the toolkit includes policy-makers, programme managers, implementers and other decision-makers involved in the planning and management of maternal and child health services in their countries.

How to use the toolkit

This toolkit has been designed to be used mainly under the supervision of the Ministry of Health, by a “country team” appointed to oversee the national guideline adaptation process. The team would include representatives of national-, district- and facility-level management, which may include programme managers from across different programmes (e.g. nutrition, HIV, tuberculosis), as well as health promotion, health management information system (HMIS) and supply chain managers, supported by implementation partners and relevant stakeholders, depending on the country context and preferences. The process of applying the toolkit would generally be led by the maternal health or safe motherhood focal person, in conjunction with a newborn health focal person (if the team is adapting the PNC recommendations). While different groups of country team members may work on different parts of the process, the findings and outputs should be regularly fed back to the broader team for review and follow-up. Maternal and newborn health national working groups may already exist and may support the process, which is described in greater detail in this user guide.

Alternatively, the toolkit can be used by teams at subnational level, or teams from other organizations, such as nongovernmental

organizations (NGOs) and professional societies, if they are actively involved in providing maternal and child health services in the country. Ideally, such teams would collaborate with the national ministry of health. For brevity, however, this guide will refer to “country teams”.

A collaborative and interdisciplinary approach that includes all relevant stakeholders in the activities outlined in the toolkit is key to ensuring the best possible outcomes. The team will have to decide whether to use this toolkit for reviewing ANC and PNC recommendations at the same time, or whether to review them separately. The same process is proposed, whether for ANC or PNC; therefore the process described in this guide can be used for both.

Introducing the four steps for adapting the WHO recommendations using the toolkit

To facilitate use of the toolkit, this user guide proposes four steps (summarized in Fig. 1), beginning with a baseline assessment of current service provision and practice (Step 1). This is followed by a stakeholder meeting (Step 2), and the creation or updating of national guidelines for ANC and/or PNC (Step 3). The process concludes with a validation meeting (Step 4).

Fig. 1. The four steps for adapting the WHO ANC and PNC recommendations

	Step 1	Step 2	Step 3	Step 4
Actions	 Assess current national ANC and/or PNC provision	 Organize a stakeholder meeting	 Create or update national ANC and/or PNC recommendations	 Organize a validation meeting
Toolkit components	User guide BAT	User guide Supplementary materials QES slides	User guide Supplementary materials	User guide BAT Supplementary materials QES slides
Toolkit outputs	Output 1 Output 2	Draft package of ANC and/or PNC interventions	Draft national ANC and/or PNC recommendations	Approved national ANC and/or PNC recommendations Plan for implementation

ANC: antenatal care; BAT: baseline assessment tool; PNC: postnatal care; QES: qualitative evidence synthesis.

Step 1

Step 1 is to conduct a baseline assessment. This is done by using the baseline assessment tool (BAT), which is available in [Web Annex A](#). The BAT will enable country teams to assess to what extent ANC and/or PNC provision and practice in country (or at subnational level) are in line with the current WHO ANC and PNC recommendations (Annexes 1 and 2). Working through the BAT worksheets, teams will produce two outputs, which are key to the baseline assessment and to the next steps in the adaptation process:

- Output 1 – a package of ANC and/or PNC interventions; and

- Output 2 – a SWOT (strengths, weaknesses, opportunities and threats) analysis of each of the recommendations and of how essential good clinical practices would be scaled up in country.

Step 2

Step 2 is to organize and carry out a stakeholder meeting to share the WHO recommendations on ANC or PNC, discuss preliminary findings from the baseline assessment, draft the package of ANC and/or PNC interventions, and initiate plans for the implementation of the updated national guidelines. Within this toolkit, users will find supplementary materials aimed at facilitating the organization and conduct of the

stakeholder meetings and developing a national ANC and/or PNC guideline ([Web Annex B](#)). The qualitative evidence synthesis (QES) slides ([Web Annex C](#)) can be used to present the values and preferences of women, their partners and their families, as well as those of health workers, which helped inform the WHO ANC and PNC recommendations. Additionally, the BAT worksheets can also be used at this stage to carry out specific exercises aimed at supporting the development of the national guidelines.

Step 3



Step 3 is to create or update national guidelines for ANC and/or PNC. To assist in this process, this toolkit provides examples of tables of contents for such guidelines ([Web Annex B](#)), as well as examples of country-specific packages of ANC and PNC interventions (Annex 3).

Step 4



Step 4 is to organize a validation meeting with the same stakeholders or with additional stakeholders, to achieve national-level consensus and approval of the guideline(s) and finalize plans for implementation.

The four steps described above have been designed to help guide thinking and decision-making at each stage of the adaptation process, and to enable country teams to choose and/or develop strategies that are likely to work in their own context. Nonetheless, it is important to recognize that the steps described in this toolkit might not happen in a linear or sequential manner. Country teams employing the toolkit might need to revisit any of the steps at any point in the process (e.g. if they find that addressing one barrier identifies another barrier) or might prefer to conduct the steps with slight variations to the suggested order.

Experiences of the country teams that pilot-tested the toolkit

Using the toolkit

There is no “one-size-fits-all” approach to using this toolkit, and use may vary across different country contexts. Pilot-testing experiences have shown that country teams might prefer to conduct the steps discussed above at different times to the ones suggested in this guide. For example, the team that used the toolkit in Rwanda completed Output 1 and Output 2 during the stakeholder meeting. This allowed for in-depth discussions in the presence of a larger number of stakeholders.



Step 1. Conducting a baseline assessment

The baseline assessment serves to evaluate to what extent current ANC and/or PNC provision and practice in a country are in line with the most recent WHO ANC and PNC recommendations (1, 3). It can also help plan the process of implementing the recommendations.

The baseline assessment is conducted by using the baseline assessment tool (BAT), which consists of 12 worksheets and is designed to help the country team evaluate the country's current antenatal and/or postnatal service provision and practice, and to plan for the implementation of the WHO recommendations. The BAT worksheets are usually completed by one or more members of the country team, typically under the direct supervision of the maternal health or safe motherhood programme manager – together with the newborn health programme manager/ focal point if PNC is also being considered – and in collaboration with other programme managers within the ministry of health and with other relevant stakeholders involved in planning and managing maternal and newborn health services, depending on country context and preferences. For example, when completing the worksheets, the team should include, as relevant, focal persons for malaria, tuberculosis, HIV, immunizations, newborn and adolescent health, and health systems.

Regardless of which team members complete the various BAT worksheets, support must be sought from policy-makers at the highest level.

It takes approximately 1–2 months to conduct the baseline assessment using the BAT worksheets. Once the assessment is complete, a report summarizing its findings is created and internally verified by the national ministry of health, before being shared at a meeting with relevant ANC and/or PNC stakeholders (see Step 2).

Contents of the baseline assessment tool (BAT)

The BAT contains 12 worksheets as detailed in Box 1. In summary, the tool begins with two sets of parallel worksheets for assessing ANC (Worksheets 1–4) and PNC (Worksheets 5–8), followed by two worksheets that apply to both ANC and PNC (Worksheets 9 and 10) and finally monitoring indicators for ANC (Worksheet 11) and for PNC (Worksheet 12). Crucially, the BAT enables teams to produce two outputs: (1) a package of ANC and/or PNC interventions and (2) a SWOT analysis for each of the ANC and/or PNC recommendations.

Box 1. Outline of the BAT worksheets for country-specific assessment of ANC and PNC

- | | |
|--|--|
| • Situational analysis data collection form | (Worksheet 1 for ANC; Worksheet 5 for PNC) |
| • Mapping of WHO recommendations to current policies and practices in country | (Worksheet 2 for ANC; Worksheet 6 for PNC) |
| • Output 1 – Package of interventions | (Worksheet 3 for ANC; Worksheet 7 for PNC) |
| • Output 2 – SWOT analysis: considerations for implementation and scale-up of recommendations | (Worksheet 4 for ANC; Worksheet 8 for PNC) |
| • Situational analysis data collection form – summary | (Worksheet 9 for both) |
| • Policy document mapping | (Worksheet 10 for both) |
| • Country-specific monitoring indicators | (Worksheet 11 for ANC; Worksheet 12 for PNC) |

Note: The BAT worksheets are available in Web Annex A at: <https://cdn.who.int/media/docs/default-source/reproductive-health/maternal-health/who-anc-pnc-toolkit-web-annex-a.zip>

SWOT: strengths, weaknesses, opportunities, threats.

The next sections describe how to complete each of the BAT worksheets; the guidance is the same whether the team is conducting a situational analysis of ANC or PNC provision in their country.

Situational analysis data collection form

There are two situational analysis data collection worksheets in the BAT. One is for collecting data on ANC service provision and practice at the national level (BAT Worksheet 1) and the other for collecting the equivalent information for PNC (BAT Worksheet 5).

To complete the worksheet, the person responsible should consult existing data sources. These could include the national

HMIS, Demographic and Health Surveys (DHS), Service Availability and Readiness Assessments (SARA), the Harmonized Health Facility Assessment (HHFA), the WHO Data Platform, National Health Accounts programme reports, and research. The point of consulting existing data sources is to minimize the cost and effort of completing the data collection form. It is crucial that the responsible person also liaises with programme managers across the different programmes within the national ministry of health, as well as with HMIS and supply chain managers and any additional relevant stakeholders, to ensure the data collected are accurate and up to date.

Each of the situational analysis data collection worksheets has two sections: section A (feedback or perspectives on ANC/PNC) and section B (current ANC/PNC provision).

Experiences of the country teams that pilot-tested the toolkit

Conducting a situational analysis

- Before starting data collection for the situational analysis, consider setting aside some time to assemble all team members involved in the use of the toolkit in order to review the different sections of the situational analysis and the type of data to be collected. This will help to ensure agreement on the way forward, particularly with regard to the sources of the data that will be used.
- During the data collection process, consider allocating some time to holding regular meetings with the team. Regular meetings are an opportunity to review the data collected, cross-check them across different sources and spot any inconsistencies or missing data.
- Have printed versions of the BAT worksheets available at meetings.
- Bear in mind that not all data issues need to be resolved at this stage and can be further discussed during the stakeholder meeting (see Step 2).

Section A of the data collection forms

This section will help to collate any direct feedback about ANC and PNC services from pregnant and postnatal women, their partners and families. The perspectives and experiences of health workers should also be logged.

If routine feedback is not incorporated into the HMIS system, a brief search for qualitative studies could be conducted using keywords such as “pregnancy”, “antenatal” and “postnatal” or “postpartum” or “newborn care”, either through an internet search or by contacting stakeholders associated with such research (e.g. universities, NGOs). Additionally, the reference sections of existing reviews could be explored (11, 12).

1. Leadership and governance
2. Health information systems
3. Service delivery
4. Health workforce
5. Financing
6. Access to essential medicines
7. Patient and population engagement
8. Existing model of ANC or PNC
9. Additional information.

1. Leadership and governance

This section focuses on the political structure within which ANC and/or PNC health services are provided, as well as leaving room for details of any additional actors, outside the ministry of health, and their roles. A current version of the ministry of health organigram should be attached to this section.

Section B of the data collection forms

This section is loosely structured around WHO’s health systems building blocks (13) and captures an array of data on current ANC and/or PNC provision in the country. The questions (asking mostly for yes/no or numerical responses) are grouped as follows:

2. Health information systems

This section will help teams record details of the existing data collection system (at facility and community level) and indicators used for decision-making in ANC and/or PNC programming at the national level. The first three items concern the routine data collection process at both facility level (item 2.1) and community

level (items 2.3 and 2.5), as well as the types of HMIS employed (item 2.2) and what information is routinely gathered and monitored (item 2.4). For item 2.4, indicator lists will vary across

countries. An example of the situational analysis data collection form for PNC (BAT Worksheet 5), including responses, is shown in Table 1.

Table 1. Worksheet 5, Situational analysis data collection form (PNC), item 2.4: excerpt showing sample responses

Other existing maternal and perinatal health indicators measured/monitored at country level by health management information system (HMIS) (expand table as necessary)							Regional or population ^a variation
	Statistic	Data	Date	Y	N	N/A	Describe variation, if any:
2.4.1	Percentage (%) of women receiving postnatal care in the facility for at least 24 hours after birth	71%	03/2019	Yes			11% in migrant populations
2.4.2	Proportion of newborns who had postnatal contact with a health worker within 2 days of delivery (%)	52%	2020			No data	
2.4.3	Percentage (%) of women receiving at least four postnatal contacts	12%	2017			No data	
2.4.4	HIV prevalence among women of reproductive age (%)	1.6%	2017		No		
2.4.5	Percentage (%) of PNC clients tested positive for HIV	0.6%	2018		No		

^a For example, among vulnerable populations such as migrants, adolescents, ethnic and racial minorities, Indigenous women, women with disabilities, sex workers, HIV-positive women, displaced and war-affected women, women living in underserved areas and other stigmatized and excluded populations.

Digital technologies and other innovations, such as rapid diagnostic tests, are included in items 2.6 and 2.7, respectively. Item 2.8 helps teams gain a broad overview of maternal health in the country. Most of the indicators are derived from population-based surveys, such as DHS or Multiple Indicator Cluster Surveys (MICS); others are available only from aggregate national data (e.g. case fatality rate for top maternal causes of death – item 2.8.11) or by special enquiry (e.g. facilities with basic emergency obstetric care and comprehensive

emergency obstetric care signal functions – items 2.8.12 and 2.8.13).

Item 2.9, which outlines population characteristics, is crucial for establishing the parameters within which context-specific recommendations are applied. For example, looking at the ANC worksheet, if the country is malaria endemic (item 2.9.11), or has a malaria-endemic region, the ANC recommendation in BAT Worksheet 2 (ANC recommendation mapping) for Recommendation C.6 on

intermittent preventive treatment with sulfadoxine-pyrimethamine (IPTp-SP) applies to this country (or region).

If data are not available for the questions in item 2.9, the associated recommendation, included in either BAT Worksheet 2 (ANC recommendation mapping) or BAT Worksheet 6 (PNC recommendation mapping), can be discussed during the stakeholder meeting (see Step 2). An example of this could be when the HIV prevalence in postpartum women (item 2.9.5 of the PNC situational analysis data collection form) is unknown, HIV testing in postpartum women should be included in the draft national package of PNC interventions (BAT Worksheet 7: Output 1 – PNC package) and presented to stakeholders. Lack of data for questions on population characteristics may also identify research priorities for the country. Finally, items 2.10 and 2.11 enable teams to record further details on how and how often ANC and/or PNC indicators are reviewed (item 2.11) and when they were last revised (item 2.10).

3. Service delivery

This section seeks to identify at what level of the health system ANC and/or PNC services are currently provided (i.e. community, primary, secondary or tertiary level), and to briefly describe the structure of the referral system. If any ANC and/or PNC outreach activities are conducted, they should be

documented in item 3.1.6. Items 3.6 and 3.7 in BAT Worksheet 5 (Situational analysis data collection form [PNC]) gathers information about whether the health system is organized to provide PNC contacts to mother and baby together in the first week and after the first week. This information is important, because the WHO model seeks to ensure that the mother–infant pairs are cared for together by providing attention to both at each contact.

4. Health workforce

Item 4.1 seeks to investigate the existing human resources (focusing on the public sector) and item 4.2 the status of health workers' current in-service training for ANC and/or PNC. Item 4.3 of BAT Worksheet 1 records which health worker (e.g. doctor, midwife, community health worker [CHW]) is allowed to complete which ANC-related task (e.g. taking blood pressure measurements, distributing nutritional supplements, performing ultrasound scans). The equivalent worksheet for PNC (BAT Worksheet 5) records which health worker is permitted to complete which PNC-related tasks, such as newborn screening for abnormalities of the eye or for hearing impairment, screening for common maternal mental health conditions or discharge preparedness and readiness assessment. Examples of responses to the questions in item 4.3 in BAT Worksheet 1 are shown in Table 2.

Table 2. Worksheet 1, Situational analysis data collection form (ANC), item 4.3: excerpt showing sample responses

4.3	Please indicate which cadre is allowed to perform the following antenatal care (ANC) tasks									
		Advanced level associate clinician	Associate clinician	Non-specialist doctor	Midwife	Auxiliary nurse midwife	Auxiliary nurse	Nurse	Lay health worker	Other cadre
4.3.1	Maternal weight/height measurement	X	X	X	X	X	X	X	X	
4.3.2	Blood pressure measurement	X	X	X	X	X	X			
4.3.3	Clinical estimate of gestational age	X	X	X	X	X	X			
4.3.4	Fetal heart rate (FHR) auscultation	X	X	X	X	X	X			
4.3.5	Detection of breech presentation	X	X	X	X	X	X			
4.3.6	Vaginal examination	X	X	X	X	X	X			
4.3.7	Syphilis screening	X	X	X	X	X	X			
4.3.16	Iron/folic acid supplement dispensing	X	X	X	X	X	X	X	X	X
4.3.28	Counselling on postpartum contraception	X	X	X	X	X	X	X	X	X

5. Financing

The questions in item 5 refer to financing for ANC and/or PNC services, including available insurance schemes, if any, and whether services have been costed. Item 5.4 seeks to identify which ANC and/or PNC services women pay a fee for, according to government policies.

6. Access to essential medicines

This section enables teams to log the supply chain management system, specifically which ANC- and/or PNC-related supplies are captured and monitored (for stock-outs) by district-level systems. Item 6.3 of BAT Worksheet 1 expands on which ANC-related

amenities (supplies and equipment, including for the laboratory) are required at which level of the health system. Section 6 in BAT Worksheet 5 captures the equivalent supply-chain-related information and amenities for PNC service provision. If a health facility assessment (14, 15) has recently been completed, the data can be incorporated into this section.

7. Patient and population engagement

This section gathers information about mechanisms that enable health services to engage with women and other members of the community and that enable the voices of women, their partners/parents and their families to be heard as regards the quality of care that pregnant women, mothers and babies receive. It also covers any special programmes that address health inequities (i.e. programmes for vulnerable groups) in ANC and/or PNC. Mechanisms for logging the views of health workers should also be established, to ensure that programmes are designed to consider their needs.

8. Existing model of ANC/PNC

This section refers to the country's current model of ANC and/or PNC provision. For ANC, BAT Worksheet 1 captures whether it is based on the focused ANC model or another model. The current contact schedule for ANC services should be detailed here. BAT Worksheet 5 captures whether the country's current model of PNC comprises the minimum of four contacts, and includes details about scheduling, booking and follow-up. Items 8.9 (BAT Worksheet 1) and 8.10 (BAT Worksheet 5) focus on the quality of care offered both before and after the birth.

Results from locally conducted qualitative research could also inform this section.

9. Additional information

This section can be used to add any additional information that might be relevant to the country context. For example, organigrams, charts or other useful documents can be copied and pasted into these worksheets or added as hyperlinks. These worksheets can be supplemented with, or extended to include, other information that stakeholders consider to be useful.

Mapping WHO recommendations to country policies and practices

BAT Worksheet 2 for ANC and Worksheet 6 for PNC assist users to map their country's existing policies to the WHO recommendations, and also show which recommendations apply to the specific country setting, based on the population characteristics (item 2.9 of the situational analysis data collection form, BAT Worksheet 1 for ANC and Worksheet 5 for PNC).

Users are asked to compare how current activities and related policies align with each of the ANC and/or PNC recommendations, who performs the index activity (main ANC/PNC provider or other), and whether a national policy change or update is necessary. Additionally, the Ministry of Health programme responsible for change should be identified, and a date for possible implementation should be set. An example of the mapping of WHO ANC Recommendation C1 to current ANC policies and practices in a country (BAT Worksheet 2) is presented in Table 3.

Table 3. Worksheet 2, Mapping of WHO ANC recommendations to current policies and practice in the country, Recommendation C1: excerpt showing sample responses

Guideline reference		C.1
WHO recommendation		A seven-day antibiotic regimen is recommended for all pregnant women with asymptomatic bacteriuria (ASB) to prevent persistent bacteriuria, preterm birth and low birth weight.
Type of recommendation		Recommended
Is the recommendation relevant? Does this recommendation apply to your context?		Yes
Current activity	What is the current service provided related to this recommendation?	Antibiotics for asymptomatic bacteriuria (ASB)
Current activity	Is this service given by the primary antenatal care (ANC) provider? If not, please expand.	No
Has the recommendation already been implemented?		No
Is any improvement required?		Guidelines and checklist
Responsible programme		Maternal, Neonatal and Child Health Department
Date for possible implementation		3 January 2025

Output 1: Package of ANC and/or PNC interventions

This section of the BAT will help the team to design packages of interventions for ANC (BAT Worksheet 3) and PNC (BAT Worksheet 7) which are suitable for their country. For ANC, it outlines all the recommended activities to be implemented during each of the eight ANC contacts scheduled over the course of the pregnancy (Table 4). For PNC, it contains all the recommended activities to be carried out in each of the four PNC contacts (Table 5).

Table 4. The WHO ANC model for a positive pregnancy: schedule of eight contacts

WHO ANC model
<i>First trimester</i>
Contact 1: up to 12 weeks
<i>Second trimester</i>
Contact 2: 20 weeks
Contact 3: 26 weeks
<i>Third trimester</i>
Contact 4: 30 weeks
Contact 5: 34 weeks
Contact 6: 36 weeks
Contact 7: 38 weeks
Contact 8: 40 weeks
<i>Return for delivery at 41 weeks if not given birth</i>

Table 5. The WHO PNC model for a positive postnatal experience: schedule of four contacts

WHO PNC model
<i>Contact 1</i>
Health facility birth: care in the facility for at least 24 hours after the birth Home birth: contact within 24 hours of the birth
<i>Contact 2</i>
Between 48 and 72 hours after the birth
<i>Contact 3</i>
Between 7 and 14 days after the birth
<i>Contact 4</i>
During week 6 after the birth

The activities which will take place during each of the ANC and PNC contacts are broken down into two categories: (i) health information (including counselling and advice) and (ii) medical assessment and management. The country team needs to specify the activities that should take place during each ANC and/or PNC scheduled contact, which health worker will perform each activity (e.g. CHW, midwife) and at what service-delivery/facility level the activity will be conducted (e.g. home visit, primary care facility). The worksheets also include a space for phasing, if necessary, to scale up an activity (e.g. the country may decide to phase in the implementation of the ANC ultrasound recommendation or the PNC

newborn screening recommendations, in order to procure equipment, train staff, etc.). A column for support mechanisms allows different resources (e.g. women's groups) and rights (e.g. birth companion) to be recorded to ensure that women and newborns receive the necessary support. An example of a contact description for ANC is shown in Table 6.

Completion of Output 1 is facilitated by the situational analysis data collection forms (BAT Worksheets 1 and 5) and the recommendation mapping sheets (BAT Worksheets 2 and 6). Worksheets 2 and 6 will also help to identify recommendations that have not yet been implemented and which should be incorporated into Output 1.

Annexes 1 and 2 each contain a table linking the WHO recommendations to the scheduled contacts. This will help country teams to determine which activities should take place at each contact, which will help them to populate the Output 1 worksheets. Table A1.1 in Annex 1 contains the WHO ANC recommendations mapped to the eight scheduled ANC contacts. Tables A2.1 and A2.2 in Annex 2 contain the 2022 WHO PNC recommendations mapped to the four scheduled PNC contacts.

When completed, Output 1 will form the basis of the materials for group work at the stakeholder meeting (see Step 2), where stakeholders can review and finalize drafting the packages of ANC and/or PNC interventions for their country.

Experiences of country teams that pilot-tested the toolkit

Output 1

- This guide suggests that a draft of Output 1 be prepared before the stakeholder meeting, but the country team may wish to work on Output 1 at a later stage (e.g. during the stakeholder meeting).
- Printed versions of the tables with each ANC and PNC recommendation mapped to each scheduled contact (Annexes 1 and 2 of this guide) can be kept on hand when Output 1 is being completed.

Table 6. Worksheet 3, Output 1 – package of ANC interventions: excerpt showing sample responses

Contact	Activity	By health worker	By health system level	Phasing, if any	Support mechanisms (e.g. women's groups, companions)
Contact 1 (up to 12 weeks)	1. Nutritional counselling on a healthy diet and physical activity	Auxiliary nurse midwife, nurse or midwife	Health centre	Phasing provision of these services to clinics in next 5 years	Ensure linkages with women's groups in the community (through community health workers [CHWs])
	2. Nutritional education on increasing daily energy and protein intake				
	3. Balanced energy and protein supplementation in undernourished populations				
	4. Daily iron and folic acid supplements				
	5. Calcium supplements				
	6. Vitamin A supplements				
	7. Restricting caffeine intake				
	1. Detection of gestational diabetes mellitus	Nurse or midwife	Health centre	Phasing provision of these services to clinics in next 5 years	Ensure woman is aware of her right to bring partner/family member to ANC contacts, has links to women's support groups and/or social services, where available
	2. Ultrasound scan				
	3. Enquiry on tobacco use and substance use				
	4. Clinical enquiry about the possibility of intimate partner violence (IPV)				
	5. Diagnosis of anaemia in pregnancy				
	6. Testing and counselling for HIV and screening for TB				
	7. Antibiotics for asymptomatic bacteriuria (ASB)				
	8. Tetanus toxoid vaccination				
	9. Pre-exposure prophylaxis for HIV prevention				
	10. Nausea and vomiting management (if needed)				
	11. Heartburn, low back and pelvic pain, leg cramps, constipation, varicose veins management (if needed)				
	12. Essential good clinical practices and counselling – maternal weight/height measurement, blood pressure measurement, birth preparedness, family planning counselling, blood typing and Rhesus (Rh) factor blood testing, management of pain during labour.				

Output 2: SWOT analysis – considerations for implementation and scale-up of recommendations

This section of the BAT involves conducting a SWOT analysis of the recommendations that the country will implement. BAT Worksheet 4 is used to complete a SWOT analysis for the ANC recommendations and Worksheet 8 for the PNC recommendations.

The SWOT analysis should be based on Output 1 – package of ANC/PNC interventions, ideally after stakeholders have reviewed it. When completing the analysis, all levels of implementation (e.g. health policy, health system and service provision and end-users – individual, family and community) should be considered. An example of the SWOT analysis (BAT Worksheet 4) for ANC is shown in Table 7.

Table 7. Worksheet 4, Output 2 – SWOT analysis: considerations for implementation and scale-up of WHO ANC recommendations: excerpt showing sample responses

Recommendation	Actions needed for implementation and scale-up	Strengths	Weaknesses	Opportunities	Threats	Ongoing implementation/research efforts
Group antenatal care (ANC)	To institutionalize ANC group care model nationally	Already in pilot phase Master trainers exist	Shortage of health workers Shortage of trained providers for group ANC Women are not yet confident in the service	Availability of pilot results	Culture that isolates men's involvement in maternal and newborn health Financial implications	Ongoing pilot study of group ANC
Ultrasound in pregnant women < 24 weeks' gestation	To introduce ultrasound scan at 24 weeks at health-centre level	Availability of qualified health workers to be trained Midwives/nurses trained as trainers	Qualified health workers are currently not trained for ultrasound scan	Commitment and alignment of partners and stakeholders (donor support)	Cost of equipment and maintenance	Ongoing implementation of a training package on obstetric ultrasound in three districts
ANC contact schedules	To scale up RapidSMS as a reminder tool for ANC services	Network of community health workers (CHWs) in every village	Limited technical capacity of CHWs Poor supervision Poor connectivity	Availability of telephone network across the country	High turnover of reproductive health staff	None at the moment
Task shifting components of ANC delivery	To involve health posts in delivering some ANC services (task shifting)	Availability of qualified human resources	Limited on-the-job training for health workers in health posts	Approved policies in favour of expansion of ANC services	High turnover of staff	Ongoing implementation of a task-shifting programme in one district

The WHO ANC recommendations (1) and the WHO PNC recommendations (3) each contain implementation considerations for each recommended intervention. The implementation considerations for the ANC recommendations are in Annex 4 of the ANC guideline and those for the PNC recommendations are in Web Annex 5 of the PNC guideline. Implementation considerations and the contents of the remarks sections for each recommendation, available

in their respective guidelines, may help to identify relevant SWOT and should help country teams to develop Outputs 1 and 2. Remarks highlight additional notes and observations from the experts involved in the development of the WHO ANC and PNC recommendations. General considerations for the adoption, implementation and scale-up of the ANC and PNC recommendations are summarized in Box 2.

Box 2. Considerations for the adoption, implementation and scale-up of the WHO ANC and PNC recommendations

Health policy considerations for adoption and scale-up of the ANC and PNC recommendations

- A firm government commitment to scale up and increase coverage of ANC and PNC services is needed, irrespective of social, economic, ethnic, racial or other factors. National support must be secured for the whole package of ANC and PNC interventions, not just for specific components.
- To be able to adequately ensure access to high-quality ANC and PNC, in the context of universal health coverage, strategies for raising public funding for health care will need revision. In low-income countries, donors could play a significant role in supporting the scale-up of implementation. Sponsoring mechanisms that support domestically driven processes to scale up the whole model are more likely to be helpful than mechanisms that support only a part of the packages.
- To set the policy agenda, to secure broad anchoring and to ensure progress in policy formulation and decision-making, representatives of training facilities and the relevant medical specialities and professional societies should be included in participatory processes at all stages, including prior to an actual policy decision, to secure broad support for scaling up.
- To facilitate negotiations and planning, situation-specific information on the expected impact of implementation of the ANC and PNC recommendations on service users, health workers and costs should be compiled and disseminated. Furthermore, the intervention packages should be adapted to local context and service-delivery settings.

Health system or organizational-level considerations for implementation of the ANC and PNC recommendations

- National and subnational subgroups may be established to adapt and implement these recommendations, including development or revision of existing national/subnational guidelines or protocols.
- Introduction of the ANC and PNC models should involve pre-service training institutions and professional bodies, so that the training curricula can be updated as quickly and smoothly as possible.
- Long-term planning is needed for resource generation and budget allocation to address the shortage of skilled health personnel and trained community health workers, to improve facility infrastructure and referral pathways, and to strengthen and sustain high-quality ANC and PNC services.

- In-service training and supervisory models will need to be developed according to health workers' professional requirements, considering the content, duration and procedures for the selection of health workers for training. These models can also be explicitly designed to address staff turnover, particularly in low-resource settings.
- In-service training could integrate the ANC and/or PNC models into the models for other services. Immunization services, for example, would be a good entry point both for providing PNC and for coming into contact with women and families to provide some of the other ANC and PNC interventions.
- Standardized tools will need to be developed for supervision, ensuring that supervisors are able to support and enable health workers to deliver integrated, comprehensive ANC and PNC services.
- A strategy for task shifting may need to be developed to optimize the use of human resources.
- Tools or "job aids" for implementation at the different levels of health facility care and communities will need to be developed or updated with all key information in accordance with the ANC and PNC models.
- Strategies will need to be devised to improve supply chain management according to local requirements, such as developing protocols for the procedures of obtaining and maintaining the stock of supplies, encouraging health workers to collect and monitor data on the stock levels and strengthening the provider-level coordination and follow-up of medicines and health-care supplies required for implementation of the ANC and PNC models.
- Development or revision of national guidelines and/or health facility-based protocols based on the WHO ANC and PNC recommendations is needed.
- Good-quality supervision, communication and transport links between community, primary and higher-level facilities need to be established to ensure that referral pathways are efficient.
- Successful implementation strategies should be documented and shared as examples of best practice for other implementers. Strengthening HMIS to ensure accurate and timely data collection, reporting and analysis is crucial to monitor trends and utilization of ANC and PNC services, and to track client outcomes and plan timely interventions.
- Achieving continuity of care throughout pregnancy and the intrapartum and postnatal periods while ensuring a positive experience and healthy outcomes for women and families involves seamless coordination between different services, as well as appropriate risk stratification, comprehensive tracking of high-risk pregnancies and timely referral.

End-user-level considerations for implementation of the ANC and PNC recommendations

- Community-sensitizing activities should be undertaken to disseminate information about the importance of each component of the ANC and PNC models, and on the rights of women and babies to attend ANC and receive PNC for their health and well-being. This information should provide details about the timing and content of the recommended contacts and about the expected user fees.
- It may be possible to reduce waiting times by reorganizing services and/or client flow.
- Additional activities conducted with health workers to ensure their views are taken into account will allow their workload and other systemic issues they may be facing to be considered when any changes are made.

Experiences of the country teams that pilot-tested the toolkit

Output 2

- A draft of the SWOT analysis (Output 2) can be prepared before the stakeholder meeting, or it can be tackled directly during the meeting.
- For example, one team from a pilot country completed the SWOT analysis during the stakeholder meeting. The presence of multiple stakeholders at the meeting offered the opportunity to elicit multiple considerations for the implementation and scale-up of the recommendations, from different viewpoints, ranging from health policy to health systems and end-user level.

Situational analysis: summary

BAT Worksheet 9 (Situational analysis data collection form: summary) contains all the items from Worksheet 1 (Situational analysis data collection form for ANC) and Worksheet 5 (Situational analysis data collection form for PNC). Data entered in either of these worksheets will automatically appear in the “Summary” worksheet. This will result in one sheet containing all completed items for ANC and PNC. This is especially helpful if the country team has decided to complete the BAT worksheets for adapting the ANC and PNC recommendations at the same time. Data in this summary sheet can also contribute to the situational analysis report (see Step 2).

Policy document mapping

BAT Worksheet 10 contains a policy document mapping exercise, which can be carried out as part of the baseline assessments for both ANC and PNC, or, if there are time constraints, it can be carried out later (e.g. during the validation meeting – see Step 4). A variety of international human rights, employment rights and social protection legislation exists to support women during pregnancy, and women, partners/parents and newborns in the postnatal period (16). Countries can promote an enabling environment for the implementation of the WHO ANC and PNC recommendations by ensuring they have

signed and ratified relevant treaties and drawn on human rights jurisprudence to inform ANC and/or PNC policies and programmes.

This exercise involves country teams identifying national policies and indicators relevant to the rights of the woman, the infant and the partners/parents, that are relevant to ANC and PNC and indicating the country’s progress or the status of the indicators. Useful data sources for this exercise are listed in the worksheet and include the following: the WHO Maternal, Newborn, Child and Adolescent Health and Ageing data portal (17), WHO Sexual, Reproductive, Maternal, Newborn, Child and Adolescent Health Policy Survey (18), International Labour Organization (ILO) data and statistics (19), and the Global database on the Implementation of Nutrition Action (GINA) (20).

Country-specific ANC and PNC monitoring indicators

BAT Worksheet 11 contains a table that allows country teams to draft a country-specific ANC monitoring indicators, based on the WHO ANC monitoring framework that was developed to track implementation and establish global standards for ANC (21). The process for creating the ANC framework comprised a three-pronged approach: (i) mapping of existing indicators (22), (ii) regional and global consultations for prioritization of

core and context-specific indicators and (iii) considering relevant experiences from national scale-up of the new ANC model (i.e. South Africa [23]). The core and country-specific indicators derived from the WHO ANC monitoring framework are provided in Annex 4.

BAT Worksheet 12 contains a table that allows country teams to draft country-specific PNC monitoring indicators. The core indicators included in the table are adapted from the minimum set of maternal, newborn, child and adolescent health indicators and additional indicators for routine programme monitoring included in the *Analysis and use of health facility data: guidance for maternal,*

newborn, child and adolescent health programme managers (24). The table does not include any indicators related to context-specific PNC recommendations as these are not available; nonetheless countries can decide if they will monitor these aspects and add additional indicators to the table. The PNC indicators are provided in Annex 5.

Country teams should complete the tables in BAT Worksheets 11 and 12 and use them to support plans to update their HMIS and the related indicators, in order to phase in this information for national monitoring and to feed global indicators.



Step 2. Holding a stakeholder meeting

This step aims to help the country team organize and run a stakeholder meeting to share the WHO recommendations on ANC and/or PNC, discuss preliminary findings from the situational analysis, review Outputs 1 and 2, and develop a plan for the implementation of the updated national guidelines.

The stakeholder meeting represents a valuable opportunity to bring together various actors involved in the planning, management and provision of ANC and/or PNC services in the country, review the proposed package of interventions for ANC and/or PNC, reflect on the SWOT analysis, identify any barriers and enablers, and find ways to match them with effective implementation strategies that are tailored to the country context.

Creating a report of findings from the baseline assessment

Before the stakeholder meeting, the country team will produce a report based on the baseline assessment which will include the proposed package of interventions (Output 1) and the SWOT analysis (Output 2). The

report should highlight the challenges to implementation as well as promising initiatives that could support the country's proposed new model of care. The report should be reviewed and approved by the ministry of health team before being presented at the stakeholder meeting.

It is suggested that the report should include the following sections:

1. Background
2. Baseline assessment methods
3. Key findings from the situational analysis
4. Output 1 and Output 2.

Once the stakeholder meeting has taken place, the report can be updated. It should include sections on the following:

- list of participants
- objectives of the meeting
- outcomes of the meeting
- methods used at the meeting
- outputs (including Outputs 1 and 2, finalized).

The meeting agenda and presentation materials can be included in an annex to the report.

Templates are provided to help country teams present data from the situational analysis (Supplementary material 1 in [Web Annex B](#)). These templates should be adapted as needed. For example, some of the indicators may need to be modified, depending on the data available at national level.

Drawing up the agenda and participant list

The stakeholder meeting should allow all participants to:

1. review the situational analysis
2. discuss the proposed package of interventions (Output 1); and
3. finalize the SWOT analysis (Output 2).

Countries should tailor the agenda to their needs and setting, and adapt it for use at the

stakeholder meeting. A draft agenda for the meeting is provided (Supplementary material 2 in [Web Annex B](#)).

With regard to the participants, ensuring the presence of key individuals is crucial to the success of the meeting. The country team should aim to invite representatives from different groups, in order to include a broad range of perspectives in discussions, to achieve alignment and support across the different groups, to ensure awareness of the challenges faced by each group and ultimately to help drive the implementation of the new or updated recommendations. The more key stakeholders are involved and present from the beginning of the process, the more likely they will be to facilitate and sustain the implementation of the new recommendations within the country. Box 3 sets out a list of key stakeholders to be considered when drawing up the list of participants.

Box 3. List of key stakeholders to be considered for the stakeholder meeting

This list is non-exhaustive and is intended as an example of a participant list for the stakeholder meeting. The country team organizing the meeting should draft a participant list that responds to the specific needs of the country and reflects the different groups involved in the planning, management and delivery of ANC and/or PNC.

- Representatives from the Ministry of Health, including programme managers from across different vertical programmes relevant to ANC and/or PNC (e.g. malaria, tuberculosis, immunization)
- Representatives from other relevant ministries (e.g. finance, education)
- Actors from key health systems, such as health workers (e.g. doctors, midwives, nurses, community health workers), health information workers and staff dealing with essential medicines
- Representatives from national professional associations (e.g. national midwifery, obstetrics and gynaecology, paediatrics associations)
- Representatives from United Nations agencies
- Representatives from implementing and funding partners
- Civil society and community-based representatives (e.g. representatives from women's and parents' groups, community leaders)
- Representatives from academic and/or research institutions
- Representatives from the private sector
- Any other stakeholder with influence over the ANC and/or PNC agenda within the country (e.g. health worker unions)

Using the QES slides at the meeting

The QES slides ([Web Annex C](#)) document the findings of qualitative reviews (11,12) and will be presented to set the stage for the meeting and forefront the evidence on the values and preferences of women, their partners, their families and health workers, which helped inform the development of the WHO ANC and PNC recommendations. The document is designed as a resource and not just a presentation; not all the slides will need to be presented; printed versions could be made available to stakeholders to refer to before, during or after the meeting.

Group work for the stakeholder meeting

The sections below outline the work to be done during the group work sessions at the stakeholder meeting. All the materials are provided (Supplementary materials 3, 4 and 5 in [Web Annex B](#)). There are three parts to the group work:

- Part 1: Implementation of ANC and PNC recommendations
- Part 2: Review of policy document mapping
- Part 3: Monitoring and indicators.

For a stakeholder meeting of approximately 40 participants, participants can be split into at least two groups (or another even number of groups). To ensure diversity of opinions and experience, each group should have a balanced composition of doctors, midwives and nurses, CHWs, public health officers, and representatives from NGOs, implementing partners, civil society organizations, women's groups and community organizations. Each working group should select a note-taker and

a presenter. Groups should be given at least two hours for the group work on the country's package of interventions. The presenter may then report the group's considerations and proposed modifications to the package of interventions and the SWOT analysis in the plenary session.

Part 1: Implementation of ANC and PNC recommendations

During the group work sessions, the following tasks should be completed by each group:

1. Review the package of ANC interventions (BAT Worksheet 3 – Output 1) or the package of PNC interventions (BAT Worksheet 7 – Output 1).
2. Provide inputs, comments and suggestions on the following:
 - the activities planned for each ANC/ PNC contact [WHAT?]
 - the health worker who will perform the activity (CHW, midwife, etc.) [WHO?]
 - the health-care delivery/facility level (health centre, etc.) at which the contact will take place [WHERE?]
 - the phasing plans for incorporating the activity in the medium to long term and how that will be done, for example with ultrasound or newborn screening practices [WHEN?]

Each group should be asked to review different content. For example, one group could be asked to review: (i) the content of ANC contacts 1–3 (first and second trimester), including essential good practices for each of these contacts; and (ii) the corresponding SWOT analysis for any ANC recommendations requiring a new or updated country policy (BAT Worksheet 4 – Output 2). Another group could be asked to review: (i) the contents of ANC contacts 4–8 (third trimester), including essential good practices for each of these contacts; and (ii) the corresponding SWOT analysis for any ANC recommendations

requiring a new or updated country policy (BAT Worksheet 4 – Output 2). For PNC, all group(s) could review: (i) the content of PNC contacts 1–4 for clinical assessment and management of women and newborns, including essential good practices and essential counselling and advice for maternal and newborn care,

for each of these contacts; and (ii) the SWOT analysis for any PNC recommendations requiring a new or updated country policy (BAT Worksheet 8 – Output 2). Different groups could also review the ANC and PNC recommendations in parallel, if different groups are assigned to work on these tasks.

Experiences from country teams that pilot-tested the toolkit

Planning and organizing the stakeholder meeting

To ensure the success of the meeting:

- Plan the meeting well ahead of time and discuss the participant list with all the members of the country team. Send out official invitations promptly to ensure better attendance.
- Share the meeting materials with all expected participants before the meeting (these include: the WHO ANC and/or PNC recommendations, report of findings from the situational analysis, and the QES slides; consider sharing the current national ANC and/or PNC guidelines).
- Share a draft agenda in advance of the meeting with all expected participants.
- On the day of the meeting, bring printed copies of the meeting materials and group-work materials.
- Make sure infrastructure (e.g. separate rooms or a large room, tables) and all materials for group work sessions (e.g. printed materials, projectors, screens) are available in sufficient quantities.

Part 2: Review of policy document mapping

This will involve all groups reviewing the results of the policy document mapping (BAT Worksheet 10), if this was carried out as part of the baseline assessment. The aim was

to identify relevant policies to refer to in the updated national guidelines. For each policy or convention listed, consider the current status and decide on any action to be taken (e.g. create legislation, ratify convention or use to inform ANC or PNC policy).

Experiences from country teams that pilot-tested the toolkit

Policy document mapping exercise

- It is not mandatory to carry out the review of the policy document mapping exercise during the stakeholder meeting.
- If it is not possible to complete this activity during the stakeholder meeting, organizers could plan this for the follow-up meeting to validate the updated guidelines and service packages.

Part 3: Monitoring and indicators

This will involve all groups reviewing existing indicators to monitor ANC and/or PNC and alignment with the WHO ANC and PNC indicators to ensure that ANC and/or PNC services are being delivered in accordance with the country's package of interventions (BAT Worksheets 11 and 12).

Each indicator and its source (for the numerator and denominator) should be reviewed. Indicators not currently captured routinely in HMIS should be prioritized for the next national update of data collection tools at facility level. Aggregation at national level should also be planned.



Step 3. Creating or updating national ANC and PNC guidelines

Once the package of ANC and/or PNC interventions (Output 1) has been discussed and agreed by stakeholders, a national guideline should be drafted. Possible structures for these guidelines are outlined in [Web Annex B](#) (Supplementary material 6).

Table 8 provides an example of how part of a package of ANC interventions could be detailed in the national ANC guideline. Table 9 provides an example of how part of a package of PNC interventions could be detailed in the national PNC guideline. These two tables show the first contacts only. The full tables are provided in Annex 3.

Table 8. Example of a country-specific package of ANC interventions, showing initial and first contact only

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Laboratory tests	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Initial contact regardless of gestation	Full history Clinical estimation of gestational age Exhaustive physical examination Abdominal palpation Tuberculosis (TB) screening Screening for intimate partner violence (IPV)	Refer to the gestational age	Nutrition, diet, physical activity and rest HIV and sexually transmitted infection (STI) tests Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of long-lasting insecticidal nets (LLINs) Danger signs and emergency preparedness Early stimulation of the baby Family planning Importance of attending all eight ANC contacts	HIV Syphilis Hepatitis B Urine dipstick Malaria Haemoglobin (Hb) estimation Asymptomatic bacteriuria (ASB) (Gram stain or culture) (depending on facility)	Blood group and Rhesus Full blood count	Ultrasound	Iron and folic acid (IFA) Tetanus toxoid/ tetanus diphtheria (TT/TD) vaccine Treat or advise about relief of common physiological symptoms Combination prevention: pre-exposure prophylaxis for HIV (PrEP), condom, voluntary medical male circumcision (VMMC) for men	Maternal weight/height measurement Mid-upper arm circumference (MUAC) Blood pressure measurement Male involvement STI screening	Common physiological disorders including STIs Complications of early pregnancy	Complete ANC card and give to woman Fill ANC register/ community management information system (CMIS)

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Laboratory tests	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Contact 1 at 8–12 weeks	Full history									
	Clinical estimation of gestational age		Nutrition, diet and physical activity, rest, HIV and STI test				IFA			
			Adherence to medicines prescribed	HIV	Blood group and Rhesus		TT/TD	Maternal weight/height measurement	Common physiological disorders	Complete ANC card and give to woman
	Exhaustive physical examination		Avoid caffeine intake, alcohol, tobacco and substance abuse	Syphilis	Full blood count	Ultrasound	Treat or advise about relief of common physiological symptoms	MUAC	including STIs	
	Abdominal palpation		Use of LLINs	Hepatitis B	Check for ASB (Gram stain or culture)		Combination prevention: PrEP, condom, VMMC for men	Blood pressure measurement	Complications of early pregnancy	Fill in ANC register/CMIS
	TB screening		Danger signs	Urine dipstick				Male involvement		
	Screening for IPV		Emergency preparedness	Malaria				STI screening		
			Early stimulation of the baby	Hb estimation						
		Family planning				Antibiotics for ASB				
Notes: - ANC services are offered at health posts and health centres, and district, provincial and referral hospitals. - All cadres of health workers will be expected to provide information, education and communication. All midwives will be expected to provide individual counselling. - Complete an ANC card, enter information into the register, and give the card to the pregnant woman. - Refer women with complications to the next level of care.										

Table 9. Example of a country-specific package of PNC interventions, showing first contact only

Contact	Maternal assessments	Maternal care	Counselling and advice for maternal care	Newborn assessment	Newborn care	Counselling and advice for newborn care	Point-of-care tests	Preventive measures for the woman	Preventive measures for the newborn	Essential good clinical practices	Health system/ health promotion
Contact 1 First 24h	Assessment of the woman Screening for postpartum anxiety and depression	Local cooling for perineal pain relief Oral analgesia for perineal pain relief Pharmacological relief of pain due to uterine cramping/ involution Non-pharmacological interventions to treat breast engorgement Postpartum oral iron and folate supplementation Provision of comprehensive contraceptive information and services during PNC	When to return for routine contacts Danger signs and care seeking Breastfeeding Postpartum contraception Sexual activity Prevention of malaria Postnatal hygiene for woman Postnatal nutrition and physical activity	Assessment of the newborn for danger signs Universal screening for neonatal hyperbilirubinaemia Universal screening for abnormalities of the eye, hearing impairment	Exclusive breastfeeding Protect, promote and support breastfeeding in facility Whole body massage for healthy term infants Early childhood development	When to return for routine contacts Danger signs and care seeking Breastfeeding Warmth for the baby Cord care Sleeping position Postnatal hygiene for baby, including bathing Immunization	HIV catch-up testing Hepatitis B	Non-pharmacological interventions to prevent mastitis Prevention of postpartum constipation Preventive anthelmintic and schistosomiasis treatment Oral pre-exposure prophylaxis (PrEP) for HIV prevention Prevention of postpartum anxiety and depression	Clean cord care Sleeping position for the prevention of sudden infant death syndrome (SIDS) Prevention of infections through vaccination (hepatitis B, polio, BCG)	Woman: Blood group and Rhesus Check tetanus toxoid immunization status and give vaccination if due	Midwife-led continuity of care Task shifting components of postnatal care delivery Recruitment and retention of staff in rural and remote areas Involvement of men in PNC and maternal and newborn health Home-based records



Step 4. Holding a validation meeting for stakeholders

Once the national guidelines for ANC and/or PNC have been created or updated (Step 3), the stakeholders who contributed to their development (in Steps 1 and 2) can meet again to validate them and prepare long- and short-term implementation plans. The validation meeting will help to achieve national-level consensus and approval of the guideline(s). Other relevant stakeholders may be invited at this stage of the process if they were not included in previous steps. These could be representatives from the ministry of finance or the ministry of education, or others involved in relevant policy-making, such as members of committees on social affairs and/or gender.

As part of the implementation strategy, stakeholders may wish to make plans for the printing and dissemination of the guideline(s), the development of training materials and job aids, and the development and implementation of a communication plan for dissemination among health workers. They may also want to consider advocacy strategies to obtain all the items necessary for the implementation

of the new packages (e.g. equipment, supplies, drugs and commodities). They could also plan for supervision, monitoring and evaluation, and updates. [Web Annex D](#) summarizes the updates to the 2016 WHO ANC recommendations and is intended to be revised on an ongoing basis to include further updates of the ANC or PNC recommendations.

In fact, the validation meeting may also offer an opportunity to discuss the updating of HMIS tools (e.g. facility registers, personal records/home-based records, data reporting forms) and related indicators. To this end, it might be useful to refer to the country-specific ANC and PNC monitoring indicators – BAT Worksheets 11 and 12 ([Web Annex A](#)). Additionally, if the policy document mapping exercise (BAT Worksheet 10) has not yet been carried out, it might be beneficial to use the validation meeting as an opportunity to involve all stakeholders in the process of identifying relevant policies to refer to in the updated national guidelines, or considering which ones may need to be developed or updated.

Planning for country implementation of the national ANC and/or PNC guidelines

Once the contents of the national guidelines for ANC and/or PNC have been validated and national-level consensus and approval have been obtained, country teams can consider different ways of supporting the implementation of the new or updated national guidelines.

It is outside the scope of this user guide to provide detailed guidance on the development of an implementation plan with strategies tailored to the country context, but some general implementation considerations can be found in Step 1 of this user guide under Output 2: SWOT analysis. A few practical examples from the pilot-testing of this toolkit and the experiences of countries that have begun implementing the recommendations are shared below.

Country implementation usually relies on the development of a national roadmap setting out how the country can implement the new guidelines within the context of the national strategic Maternal, Newborn and Child Health (MNCH) direction and national MNCH targets and goals. The roadmap contains strategies for guideline introduction and implementation, including details of changes to the organization, planning and delivery of ANC and/or PNC health services, and of activities for stakeholder and community engagement. These plans often also include aspects of monitoring and evaluation, including monitoring the availability of supplies (drugs, supplements, testing kits, equipment) which are required for implementing the new packages of interventions. The roadmap contains the operational plan, a practical document that identifies the processes, resources and action needed to implement the activities (25). One approach could be to start testing the operational plan on a small scale first, to be able to monitor the feasibility of the implementation, identify the type of on-site support that is required, seek feedback from participants and use that information to improve the national roadmap and operational plan before they are rolled out properly.

Experiences of country teams in developing implementation plans

During the stakeholder meeting in Sierra Leone (6) stakeholders found that the following would be needed to support the implementation of the national ANC guidelines:

- printing and dissemination of the national guideline
- review of the national Maternity Record Card
- development of training materials and job aids
- development and implementation of a communication plan for dissemination among health workers
- a plan to make available all items needed for the implementation of the ANC package (e.g. equipment, supplies, drugs, commodities)
- planning for supervision, monitoring and evaluation.

During the stakeholder meeting in Guinea, participants were asked to work in small groups to develop proposals for the implementation of the ANC and PNC packages, including a pilot for implementation at district level before national scale-up. These draft plans were then presented in plenary, and advantages and disadvantages were discussed. The revised plans will form the basis of the national roadmap that will be developed by Guinea's Ministry of Health and Public Hygiene.

To support the translation of the new or updated national ANC and PNC guidelines

into clinical practice and service delivery, new derivatives (e.g. training materials, protocols,

job aids and handbooks) will have to be developed, or existing ones updated. Where applicable, existing registries, medical records,

client cards and home-based records will have to be revised to reflect any updates to the ANC or PNC intervention packages.

Experiences of country teams in developing guideline derivatives

The WHO Regional Office for Africa developed training resources (including facilitator guides and presentations) based on the WHO ANC recommendations. These training materials can be further tailored by countries. Where appropriate, a “training of trainers” model could be applied.

Specific training activities may be needed, for example for the conduct of antenatal ultrasounds before 24 weeks of gestation, or newborn screening for hyperbilirubinaemia, eye problems or hearing, or to expand content to include management of common complications of pregnancy and postnatal periods.

In Sierra Leone, the national Maternity Record Card was reviewed and harmonized with the new ANC package of interventions (6). In Uganda, the ANC client take-home card, TB screening, and birth and emergency preparedness plan tools were reviewed to reflect the country’s updated ANC intervention package, and piloted in selected sites before national scale-up (6).

Consideration can also be given to the use of digital technologies to facilitate the implementation of ANC and PNC intervention packages. In this sense, WHO has developed the SMART – standards-based, machine-readable, adaptive, requirements-based and testable – guidelines approach to facilitate implementation through digital systems (26). Digital adaptation toolkits are developed to facilitate interpretation and then adaptation of the guideline content into digital tools (27).

The perspectives of women, their partners, families and communities should be included in the development of the implementation plans, and that of the guideline derivatives and dissemination activities. This could be done through existing women’s or parents’ groups, community leaders and CHWs. Activities to disseminate messaging regarding the

new ANC and PNC packages may include conducting community-level sensitization meetings, giving educational talks at women/parental/family support group meetings, disseminating information materials (e.g. posters, infographics, flipcharts) and conducting media campaigns (28).

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Annex 1. WHO antenatal care (ANC) model for a positive pregnancy experience: recommendations mapped to eight scheduled ANC contacts

The table below, adapted from the 2016 *WHO recommendations on antenatal care for a positive pregnancy experience (1)* (p. 108), describes which activities should take place at each contact during ANC. This table will help country teams to populate Output 1 – package of ANC interventions (Worksheet 3, available in [Web Annex A](#). Baseline assessment tool [BAT]). Please note that actions recommended only in a research context should not be included unless there are plans and funds available for the research.

Table A1.1 WHO ANC model for a positive pregnancy experience: recommendations mapped to eight scheduled ANC contacts

Type of intervention	Type of recommendation	Eight scheduled ANC contacts (weeks of gestation)							
		1	2	3	4	5	6	7	8
		(12 weeks)	(20 weeks)	(26 weeks)	(30 weeks)	(34 weeks)	(36 weeks)	(38 weeks)	(40 weeks)
A. Nutritional interventions									
A.1.1 Nutritional counselling on a healthy diet and physical activity	Recommended	X	X	X	X	X	X	X	X
A.1.2 Nutritional education on increasing daily energy and protein intake	Context-specific recommendation	X	X	X	X	X	X	X	X
A.1.3 Balanced energy and protein supplementation in undernourished populations	Context-specific recommendation	X	X	X	X	X	X	X	X
A.2.1 Daily iron and folic acid supplements	Recommended	X	X	X	X	X	X	X	X
A.2.2 Intermittent iron and folic acid supplements	Context-specific recommendation	X	X	X	X	X	X	X	X
A.3 Calcium supplements	Context-specific recommendation	X	X	X	X	X	X	X	X

Type of intervention	Type of recommendation	Eight scheduled ANC contacts (weeks of gestation)							
		1	2	3	4	5	6	7	8
		(12 weeks)	(20 weeks)	(26 weeks)	(30 weeks)	(34 weeks)	(36 weeks)	(38 weeks)	(40 weeks)
A.4 Vitamin A supplements	Context-specific recommendation	X	X	X	X	X	X	X	X
A.5 Zinc supplements	Context-specific recommendation (research)								
A.6 Multiple micronutrient supplements	Context-specific recommendation (research)								
A.10 Restricting caffeine intake	Context-specific recommendation	X	X	X	X	X	X	X	X
Essential good clinical practices and other essential counselling (see BAT Worksheet 2, rows 54–67)		X	X	X	X	X	X	X	X
B. Maternal and fetal assessment									
B.1.1 Anaemia	Context-specific recommendation	X		X			X		
B.1.2 Asymptomatic bacteriuria (ASB)	Context-specific recommendation	X		X		X			
B.1.3 Intimate partner violence (IPV)	Context-specific recommendation	X	X	X	X	X	X	X	X
B.1.4 Gestational diabetes mellitus (GDM)	Recommended	X	X	X	X	X	X	X	X
B.1.5 Tobacco use	Recommended	X	X	X	X	X	X	X	X
B.1.6 Substance use	Recommended	X	X	X	X	X	X	X	X
B.1.7 Human immunodeficiency virus (HIV) and syphilis	Recommended	X							
HIV catch-up testing^a	Recommended				X				
Hepatitis B surface antigen serological testing^b	Recommended	X							
B.1.8 Tuberculosis (TB)	Context-specific recommendation	X							

Type of intervention	Type of recommendation	Eight scheduled ANC contacts (weeks of gestation)							
		1	2	3	4	5	6	7	8
		(12 weeks)	(20 weeks)	(26 weeks)	(30 weeks)	(34 weeks)	(36 weeks)	(38 weeks)	(40 weeks)
B.2.1 Daily fetal movement counting	Context-specific recommendation (research)								
B.2.2 Symphysis-fundal height (SFH) measurement or abdominal palpation	Context-specific recommendation		X	X	X	X	X	X	X
B.2.4 Ultrasound scan^c	Recommended	X	X						
C. Preventive measures									
C.1 Antibiotics for asymptomatic bacteriuria (ASB)	Recommended	X		X		X			
C.2 Antibiotic prophylaxis to prevent recurrent urinary tract infections	Context-specific recommendation (research)								
C.4 Preventive anthelmintic treatment	Context-specific recommendation		X						
Preventive schistosomiasis treatment^d	Context-specific recommendation		X						
C.5 Tetanus toxoid vaccination	Recommended	X							
C.6 Malaria prevention: intermittent preventive treatment in pregnancy (IPTp)	Context-specific recommendation	X (13 weeks)	X	X	X		X		X
C.7 Pre-exposure prophylaxis for HIV prevention	Context-specific recommendation	X							
D. Interventions for common physiological symptoms									
D.1 Nausea and vomiting	Recommended	X	X	X					
D.2–6 Heartburn, leg cramps, low back and pelvic pain, constipation, varicose veins and oedema	Recommended	X	X	X	X	X	X	X	X

^a In high HIV burden settings, retest all pregnant women with unknown or HIV-negative status in late pregnancy – at third trimester visit. If either the first test or retest is missed or delayed, catch-up testing is needed. In low HIV burden settings, retest pregnant women with unknown or HIV negative status who are in serodiscordant relationships, where the partner is not virally suppressed on ART, or have other ongoing HIV risks in late pregnancy – at a third trimester visit. If either the first test or retest is missed or delayed, catch-up testing is needed (2).

- ^b In settings with a $\geq 2\%$ or $\geq 5\%$ HBsAg seroprevalence in the general population, it is recommended that HBsAg serological testing be routinely offered to all pregnant women in antenatal clinics, with linkage to prevention, care and treatment services. Couples and partners in antenatal care settings should be offered HBV testing services (3).
- ^c One ultrasound scan before 24 weeks of gestation is recommended for pregnant women to estimate gestational age, improve detection of fetal anomalies and multiple pregnancies, reduce induction of labour for post-term pregnancy, and improve a woman's pregnancy experience (4).
- ^d In endemic communities with *Schistosoma* spp. prevalence of 10% or higher, WHO recommends annual preventive chemotherapy with praziquantel in a single dose at $\geq 75\%$ up to 100% coverage in pregnant women after the first trimester, and non-pregnant adolescent girls and women of reproductive age, including women postpartum and lactating women, to control schistosomiasis morbidity and towards eliminating the disease as a public health problem. In endemic communities with *Schistosoma* spp. prevalence below 10%, WHO suggests one of two approaches based on the programmes' objectives and resources: (i) where there has been a programme of regular preventive chemotherapy, continuing preventive chemotherapy at the same or reduced frequency towards interruption of transmission; (ii) where there has not been a programme of regular preventive chemotherapy, a clinical approach of test-and-treat, instead of preventive chemotherapy targeting a population (5, 6).

References for Annex 1

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Annex 2. WHO postnatal care (PNC) model for a positive postnatal experience: recommendations mapped to four scheduled PNC contacts

The tables below, adapted from the 2022 *WHO recommendations on maternal and newborn care for a positive postnatal experience* (1), describe which activities should take place at each contact during PNC. Table A2.1 is dedicated to maternal care, while Table A2.2 covers newborn care. These tables will help country teams to determine which activities

should take place at which contact to populate Output 1 – package of PNC interventions (Worksheet 7, available in [Web Annex A](#). Baseline assessment tool [BAT]). Please note that actions recommended only in a research context should not be included unless there are plans and funds available for the research.

Table A2.1 WHO PNC model for a positive postnatal experience: recommendations mapped to four scheduled PNC contacts – maternal care

Type of intervention A. Maternal care	Type of recommendation	Four scheduled PNC contacts			
		1	2	3	4
		First 24 hours postpartum	48–72 hours postpartum	7–14 days postpartum	6 weeks postpartum
A.1 Maternal assessment					
A.1.1 Assessment of the woman	Recommended	X	X	X	X
A.1.2 Catch up HIV testing ^a	Context-specific recommendation	X			
A.1.3 Screening for TB disease ^b	Context-specific recommendation				
A.2 Interventions for common physiological signs and symptoms					
A.2.1 Local cooling for perineal pain relief	Recommended	X	X		
A.2.2 Oral analgesia for perineal pain relief	Recommended	X	X		
A.2.3 Pharmacological relief of pain due to uterine cramping/involution	Recommended	X	X		
A.2.5 Non-pharmacological interventions to treat breast engorgement	Recommended	X	X	X	X

Type of intervention	Type of recommendation	Four scheduled PNC contacts			
		1	2	3	4
A. Maternal care		First 24 hours postpartum	48–72 hours postpartum	7–14 days postpartum	6 weeks postpartum
A.3 Preventive measures					
A.3.1 Non-pharmacological interventions to prevent mastitis	Recommended	X	X	X	X
A.3.3 Prevention of postpartum constipation	Recommended	X	X	X	X
A.3.5 Preventive anthelmintic treatment^c	Context-specific recommendation	X (Check in 1st contact)			
A.3.6 Preventive schistosomiasis treatment^d	Context-specific recommendation	X (Check in 1st contact)			
A.3.7 Oral pre-exposure prophylaxis (PrEP) for HIV prevention	Context-specific recommendation	X			
A.4 Mental health interventions					
A.4.1 Screening for postpartum anxiety and depression	Recommended	X			
A.4.2 Prevention of postpartum anxiety and depression	Recommended	X	X	X	X
A.5 Nutritional interventions and physical activity					
A.5.1 Postpartum oral iron and folate supplementation^e	Context-specific recommendation	X	X	X	X
A.5.3 Physical activity	Recommended	X	X	X	X
A.6 Contraception					
A.6.1 Provision of contraceptive information and services	Recommended	X	X	X	X (Ongoing)
Essential good clinical practices and essential counselling and advice for maternal care (see BAT Worksheet 6, rows 10–25)		X	X	X	X

- a In high HIV burden settings, catch-up HIV testing postpartum is needed for women of HIV-negative or unknown status who missed early antenatal contact test or retesting in late pregnancy at a third trimester visit. In low HIV burden settings, catch-up HIV testing postpartum can be considered for women of HIV-negative or unknown status who missed early antenatal contact test or retesting in late pregnancy at a third trimester visit as part of the effort to eliminate mother-to-child transmission of HIV. Countries could consider this only for women who are in serodiscordant relationships, where the partner is not virally suppressed on ART, or have other known ongoing HIV risk in late pregnancy – at a third trimester visit (2).
- b (i) Systematic screening for TB disease may be conducted among the general population, including in women in the postpartum period in areas with an estimated TB prevalence of 0.5% or higher. (ii) In settings where the TB disease prevalence in the general population is 100/100 000 population or higher, systematic screening for TB disease may be conducted among women postpartum as part of postnatal care. (iii) Household contacts and other close contacts of individuals with TB disease, including women in the postpartum period and newborns, should be systematically screened for TB disease (3).
- c Depending on when last dose was administered. Preventive chemotherapy (deworming), using annual or biannual a single-dose albendazole (400 mg) or mebendazole (500 mg), is recommended as a public health intervention for all

non-pregnant adolescent girls and women of reproductive age, including women postpartum and lactating women, living in areas where the baseline prevalence of any soil-transmitted helminth infection is 20% or more among adolescent girls and women of reproductive age, in order to reduce the worm burden of soil-transmitted helminths (4).

- ^d Depending on when last dose was administered. In endemic communities with *Schistosoma* spp. prevalence of 10% or higher, WHO recommends annual preventive chemotherapy with praziquantel in a single dose at $\geq 75\%$ up to 100% coverage in pregnant women after the first trimester, and non-pregnant adolescent girls and women of reproductive age, including women postpartum and lactating women, to control schistosomiasis morbidity and towards eliminating the disease as a public health problem. In endemic communities with *Schistosoma* spp. prevalence below 10%, WHO suggests one of two approaches based on the programmes' objectives and resources: (i) where there has been a programme of regular preventive chemotherapy, continuing preventive chemotherapy at the same or reduced frequency towards interruption of transmission; (ii) where there has not been a programme of regular preventive chemotherapy, a clinical test-and-treat approach is suggested instead of preventive chemotherapy targeting a population (5).
- ^e Oral iron supplementation, either alone or in combination with folic acid supplementation, may be provided to postpartum women for 6–12 weeks following childbirth for reducing the risk of anaemia in settings where gestational anaemia is of public health concern. WHO considers a 20% or higher population prevalence of gestational anaemia to be a moderate public health problem (6).

Table A2.2 WHO PNC model for a positive postnatal experience: recommendations mapped to four scheduled PNC contacts – newborn care

Type of intervention B. Newborn care	Type of recommendation	Four scheduled PNC contacts			
		1	2	3	4
		First 24 hours after birth	48–72 hours after birth	7–14 days after birth	6 weeks after birth
B.1 Newborn assessment					
B.1.1 Assessment of the newborn danger signs	Recommended	X	X	X	X
B.1.2 Universal screening for abnormalities of the eye ^a	Recommended	X			
B.1.3 Universal screening for hearing impairment ^b	Recommended	X			
B.1.4 Universal screening for hyperbilirubinaemia ^c	Recommended	X			
B.2 Preventive measures					
B.2.1 Timing of first bath to prevent hypothermia ^d	Recommended		X		
B.2.3 Clean, dry umbilical cord care ^e	Recommended	X	X		
B.2.4 Sleeping position for the prevention of sudden infant death syndrome (SIDS)	Recommended	X	X	X	X (During the first year)
B.2.5 Newborn immunization as per latest WHO recommendations ^f	Recommended	X			
Prevention of ophthalmia neonatorum ^g	Recommended	X			
Haemorrhagic disease prophylaxis ^h	Recommended	X			

Type of intervention	Type of recommendation	Four scheduled PNC contacts			
		1	2	3	4
B. Newborn care		First 24 hours after birth	48–72 hours after birth	7–14 days after birth	6 weeks after birth
B.3 Nutritional interventions					
B.3.1 Neonatal vitamin A supplementationⁱ	Context-specific recommendation		X		
B.3.2 Neonatal vitamin D supplementation	Context-specific recommendation (research)				
B.4 Infant growth and development					
B.4.1 Whole body massage for healthy term newborns	Recommended	X	X	X	X
B.4.2 Early childhood development^j	Recommended	X	X	X	X
B.5 Breastfeeding					
B.5.1 Exclusive breastfeeding^k	Recommended	X	X	X	X (Until 6 months of age)
B.5.2 Protecting, promoting and supporting breastfeeding in facilities providing maternity services^l	Recommended	X	X	X	X
Essential counselling and advice for newborn care (see BAT Worksheet 6 for specific activities, rows 26–34)		X	X	X	X

- ^a Universal newborn screening for abnormalities of the eye should be done prior to discharge after facility birth, and at first contact in the healthcare facility for home birth, ideally within first six weeks after birth. Newborn eye screening should be integrated with the existing postnatal care services. For example, to be included in the discharge preparedness checklist after facility birth and as part of newborn examination in well-baby clinics in the first year after birth.
- ^b Universal screening for hearing impairment can be performed in the hospital soon after birth but also in the well-baby clinic and community health clinics in the first month after birth. Settings must be quiet environments, so the baby is settled and ideally sleeping before the screen is performed.
- ^c Universal transcutaneous bilirubinometer (TcB) screening should be part of the discharge readiness assessment for the newborn and can be done at the time of routine newborn discharge examination.
- ^d The first bath of a term healthy newborn should be delayed for more than 24 hours after birth - and can take place either in the facility or at home.
- ^e Clean, dry, umbilical cord care is recommended. Daily application of 4% chlorhexidine (7.1% chlorhexidine digluconate aqueous solution or gel, delivering 4% chlorhexidine) to the umbilical cord stump in first week after birth is recommended only in settings where harmful traditional substances (such as animal dung) are commonly used on the umbilical cord.
- ^f All infants should receive the first dose of the hepatitis B vaccine as soon as possible after birth, ideally within 24 hours. If administration within 24 hours is not feasible, all infants should receive the birth dose during the first contact with health facilities, up to the time of the first primary dose series. In polio-endemic countries and countries at high risk for importation and subsequent spread, WHO recommends a bivalent oral polio vaccination (bOPV) birth dose followed by a primary series of three bOPV doses and at least one inactivated poliovirus dose. In countries or settings with a high incidence of TB disease and/or leprosy, a single dose of BCG vaccine should be given to neonates at birth, or as soon as possible thereafter, for prevention of TB and leprosy. If it cannot be given at birth, it should be given at the earliest opportunity thereafter and should not be delayed (7).

- ^g The infant's eyes should be carefully cleaned immediately after birth. The application of 1% silver nitrate solution or 1% tetracycline ointment to the eyes of all infants at the time of delivery is strongly recommended as a prophylactic measure. If missed, give before 24 hours (8).
- ^h All newborns should be given 1 mg of vitamin K intramuscularly after birth (i.e. after the first hour by which the infant should be in skin-to-skin contact with the mother and breastfeeding should be initiated). If missed, give before 24 hours (9).
- ⁱ Routine neonatal vitamin A supplementation is not recommended to reduce neonatal and infant mortality in all settings. In settings with recent (within the last five years) and reliable data that indicate a high infant mortality rate (greater than 50 per 1000 live births) and high prevalence of maternal vitamin A deficiency (> 10% of pregnant women with serum retinol concentrations < 0.70 µmol/L), providing newborns with a single oral dose of 50 000 IU of vitamin A within the first three days after birth may be considered to reduce infant mortality.
- ^j All infants and children should receive responsive care between 0 and 3 years of age; parents and other caregivers should be supported to provide responsive care. All infants and children should have early learning activities with their parents and other caregivers between 0 and 3 years of age; parents and other caregivers should be supported to engage in early learning with their infants and children. Support for responsive care and early learning should be included as part of interventions for optimal nutrition of infants and young children. Psychosocial interventions to support maternal mental health should be integrated into early childhood health and development services (10).
- ^k All babies should be exclusively breastfed from birth until 6 months of age. Mothers should be counselled and provided support for exclusive breastfeeding at each postnatal contact (11).
- ^l Facilities providing maternity and newborn services should have a clearly written breastfeeding policy that is routinely communicated to staff and parents. Health-facility staff who provide infant feeding services, including breastfeeding support, should have sufficient knowledge, competence and skills to support women to breastfeed. Key clinical practices women and newborns should receive to successfully establish and maintain breastfeeding:
 - Discuss the importance and management of breastfeeding with pregnant women and their families.
 - Facilitate immediate and uninterrupted skin-to-skin contact and support mothers to initiate breastfeeding as soon as possible after birth.
 - Support mothers to initiate and maintain breastfeeding and manage common difficulties.
 - Do not provide breastfed newborns any food or fluids other than breastmilk, unless medically indicated.
 - Enable mothers and their infants to remain together and to practise rooming-in 24 hours a day.
 - Support mothers to recognize and respond to their infants' cues for feeding.
 - Counsel mothers on the use and risks of feeding bottles, teats and pacifiers.
 - Coordinate discharge so that parents and their infants have timely access to ongoing support and care (12).

References for Annex 2

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Annex 3. Examples of country-specific antenatal care (ANC) and postnatal care (PNC) intervention packages

These tables provide examples of how a package of ANC or PNC interventions (i.e. Output 1) could be detailed in the national ANC or PNC guideline (see Step 3 in this guide).

Table A3.1 Example of national package of ANC interventions

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Labs	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Initial contact regardless of gestation	Full history Clinical estimation of gestational age Exhaustive physical examination Abdominal palpation Tuberculosis (TB) screening Screening for intimate partner violence (IPV)	Refer to the gestational age	Nutrition, diet, physical activity and rest HIV and sexually transmitted infection (STI) tests Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of long-lasting insecticidal nets (LLINs) Danger signs and emergency preparedness Early stimulation of the baby Family planning Importance of attending all eight ANC contacts	HIV Syphilis Hepatitis B Urine dipstick Malaria Haemoglobin (Hb) estimation	Blood group and Rhesus Full blood count Asymptomatic bacteriuria (ASB) (Gram stain or culture, depending on facility)	Ultrasound	IFA Tetanus toxoid/tetanus diphtheria (TT/TD) vaccine Treat or advise about relief of common physiological symptoms Combination prevention: pre-exposure prophylaxis for HIV (PrEP), condom, voluntary medical male circumcision (VMMC) for men	Maternal weight/height measurement Mid-upper arm circumference (MUAC) Blood pressure measurement Male involvement STI screening	Common physiological disorders including STIs Complications of early pregnancy	Complete ANC card and give to woman Fill ANC register/ community management information system (CMIS)

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Labs	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Contact 1 at 8–12 weeks	Full history Clinical estimation of gestational age Exhaustive physical examination Abdominal palpation TB screening Screening for IPV		Nutrition, diet and physical activity, rest, HIV and STI test Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness Early stimulation of the baby Family planning	HIV Syphilis Hepatitis B Urine dipstick Malaria Hb estimation	Blood group and Rhesus Full blood count ASB	Ultrasound	IFA TT/TD Treat or advise about relief of common physiological symptoms Combination prevention: PrEP, condom, VMMC for men Antibiotics for ASB	Maternal weight/height measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders including STIs Complications of early pregnancy	Complete ANC card and give to woman Fill in ANC register/CMIS
Contact 2 at 20 weeks	Ask the woman if she has experienced any problems/changes since last visit Clinical estimation of gestational age Physical examination Symphysis-fundal height measurement/ abdominal palpation TB screening Screening for IPV	Fetal heart rate (FHR)/ fetal heart sound auscultation after 20 weeks Ask about fetal movement	Nutrition, diet, physical activity and rest HIV and STI test Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness Early stimulation of the baby Family planning	Urine dipstick, HIV, syphilis, retest for those who were negative or not tested in the first contact Hepatitis B Malaria		Ultrasound if not yet done	Anthelmintic (single-dose mebendazole 500 mg) IFA Calcium Combination prevention: PrEP, condom, VMMC for men	Maternal weight measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders including STIs Complications of pregnancy	Complete ANC card and give to woman Fill in ANC register/CMIS

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Labs	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Contact 3 at 26 weeks	Ask the woman if she has experienced any problems/changes since last visit Clinical estimation of gestational age Physical examination Symphysis-fundal height measurement/ abdominal palpation TB screening Screening for IPV Rule out PROM	FHR auscultation Ask about fetal movements	Nutrition, diet, physical activity, and rest HIV and STI test Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness Early stimulation of the baby Family planning	Urine dipstick Malaria	Hb ASB	Ultrasound if indicated	IFA Calcium Combination prevention: PrEP, condom, VMMC for men	Maternal weight measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders including STIs Complications of pregnancy	Complete ANC card and give to woman Fill ANC register/CMIS
Contact 4 at 30 weeks	Ask the woman if she has experienced any problems/changes since last visit Clinical estimation of gestational age Physical examination Symphysis-fundal height measurement/ abdominal palpation TB screening Screening for IPV Ask about fetal movement Rule out PROM	FHR auscultation Ask about fetal movements	Nutrition, diet physical activity, and rest HIV and STI test Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness Early stimulation of the baby Family planning	Urine dipstick HIV and syphilis re-test for those who were negative Malaria		Ultrasound if indicated	IFA Calcium Combination prevention: PrEP, condom, VMMC for men	Maternal weight measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders Manage complications of pregnancy	Complete ANC card and give to woman Fill ANC register/CMIS

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Labs	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Contact 5 at 34 weeks	Ask the woman if she has experienced any problems/changes since last visit Clinical estimation of gestational age Physical examination Symphysis fundal height measurement/ abdominal palpation TB screening Screening for IPV Rule out PROM	FHR auscultation Ask about fetal movements	Nutrition, diet and physical activity, rest, HIV and STI test Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness Early stimulation of the baby Family planning	Urine dipstick Malaria	ASB	Ultrasound if indicated	IFA, calcium, combination prevention: PrEP, condom, VMMC for men	Maternal weight measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders Manage complications of pregnancy	Complete ANC card and give to woman Fill ANC register/CMIS
Contact 6 at 36 weeks	Ask the woman if she has experienced any problems/changes since last visit Clinical estimation of gestational age Physical examination Symphysis-fundal height measurement/ abdominal palpation TB screening Screening for IPV Rule out PROM	FHR auscultation Ask about fetal movements	Nutrition, diet, physical activity, and rest HIV and STI test Adherence to medicines prescribed Avoid caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness Early stimulation of the baby Family planning	Urine dipstick Malaria	Hb	Ultrasound if indicated	IFA Calcium Combination prevention: PrEP, condom, VMMC for men	Maternal weight measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders Manage complications of pregnancy	Complete ANC card and give to woman Fill ANC register/CMIS

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Labs	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Contact 7 at 38 weeks	Ask the woman if she has experienced any problems/changes since last visit Clinical estimation of gestational age Physical examination Symphysis fundal height measurement/ abdominal palpation TB screening Screening for IPV Rule out PROM	FHR auscultation Ask about fetal movements	Nutrition, diet, physical activity, and rest HIV test Avoid: caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness Infant and young child feeding (IYCF) Breastfeeding Breast care Family planning /healthy timing and spacing of pregnancy (HTSP) Birth planning Early infant male circumcision (EIMC) Early stimulation of the baby	Urine dipstick HIV and Syphilis re-test for those who were negative Malaria		Ultrasound if indicated	IFA Calcium Combination prevention: PrEP, condom, VMMC for men	Maternal weight measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders Manage complications of pregnancy	Complete ANC card and give to woman Fill ANC register/CMIS

Contact	Maternal assessment	Fetal assessment	Counselling	Point-of-care tests	Labs	Radiological tests	Preventive measures	Essential good clinical practices	Treat/ manage	Others
Contact 8 at 40 weeks	Ask the woman if she has experienced any problems/changes since last visit Clinical estimation of gestational age Physical examination Symphysis fundal height measurement/ abdominal palpation TB screening Screening for IPV Rule out PROM Ask about symptoms of labour	FHR auscultation Ask about fetal movements	Nutrition, diet, physical activity, and rest HIV test Avoid: caffeine intake, alcohol, tobacco and substance abuse Use of LLINs Danger signs Emergency preparedness IYCF Breastfeeding Breast care Family Planning/HTSP Birth planning EIMC Early stimulation of the baby Counsel on admission if no signs of labour at 41 weeks	Urine dipstick Malaria		Ultrasound if indicated	IFA Calcium Combination prevention: PrEP, condom VMMC for men	Maternal weight measurement MUAC Blood pressure measurement Male involvement STI screening	Common physiological disorders Manage complications of pregnancy	Complete ANC card and give to woman Fill ANC register/CMIS
Health worker to provide the services	Midwives, obstetricians, doctors	Midwives, obstetricians, doctors	Midwife, nurses, dieticians, nutritionists, health promotion, community health workers (CHWs), doctors, obstetricians	Midwives, nurses, doctors	Lab technician	Radiology, obstetricians, midwives, sonographers (if any)	Nurses and midwives	Midwives, (blood pressure – BP), obstetricians (BP), doctors (BP); CHWs (weight, MUAC, height)	Midwives, doctors, obstetricians	Midwives, obstetricians, doctors

Notes:

- ANC services are offered at health posts and health centres, and district, provincial and referral hospitals.
- All cadres of health workers will be expected to provide information, education and communication. All midwives will be expected to provide individual counselling.
- Complete an ANC card, enter information into the register, and give the card to the pregnant woman.
- Refer women with complications to the next level of care.

Table A3.2 Example of package of PNC interventions

Contact	Maternal assessments	Maternal care	Counselling and advice for maternal care	Newborn assessment	Newborn care	Counselling and advice for newborn care	Point-of-care tests	Preventive measures for the woman	Preventive measures for the newborn	Essential good clinical practices	Health system/ health promotion
Contact 1 First 24 hours	Assessment of the woman Screening for postpartum anxiety and depression	Local cooling for perineal pain relief Oral analgesia for perineal pain relief Pharmacological relief of pain due to uterine cramping/ involution Non-pharmacological interventions to treat breast engorgement Postpartum oral iron and folate supplementation Provision of comprehensive contraceptive information and services during PNC	When to return for routine contacts Danger signs and care seeking Breastfeeding Postpartum contraception Sexual activity Prevention of malaria Postnatal hygiene for woman Postnatal nutrition and physical activity	Assessment of the newborn for danger signs Universal screening for neonatal hyperbilirubinaemia Universal screening for abnormalities of the eye, hearing impairment	Exclusive breastfeeding Protect, promote and support breastfeeding in facility Whole body massage for healthy term infants Early childhood development	When to return for routine contacts Danger signs and care seeking Breastfeeding Warmth for the baby Cord care Sleeping position Postnatal hygiene for baby, including bathing Immunization	HIV catch-up testing Hepatitis B	Non-pharmacological interventions to prevent mastitis Prevention of postpartum constipation Preventive anthelmintic and schistosomiasis treatment Oral pre-exposure prophylaxis (PrEP) for HIV prevention Prevention of postpartum anxiety and depression	Clean cord care Sleeping position for the prevention of sudden infant death syndrome (SIDS) Prevention of infections through vaccination (hepatitis B, polio, BCG)	Woman: Blood group and Rhesus Check tetanus toxoid immunization status and give vaccination if due	Midwife-led continuity of care Task shifting components of postnatal care delivery Recruitment and retention of staff in rural and remote areas Involvement of men in PNC and maternal and newborn health Home-based records

Contact	Maternal assessments	Maternal care	Counselling and advice for maternal care	Newborn assessment	Newborn care	Counselling and advice for newborn care	Point-of-care tests	Preventive measures for the woman	Preventive measures for the newborn	Essential good clinical practices	Health system/ health promotion
Contact 2 48–72 hours	Assessment of the woman Screening for postpartum anxiety and depression	Local cooling for perineal pain relief Oral analgesia for perineal pain relief Pharmacological relief of pain due to uterine cramping/ involution Non-pharmacological interventions to treat breast engorgement Postpartum oral iron and folate supplementation Provision of comprehensive contraceptive information and services during PNC	When to return for routine contacts Danger signs and care seeking Lactation Postpartum contraception Sexual activity Prevention of HIV and sexually transmitted infections (STIs) Prevention of malaria Tobacco and other substance use Postnatal hygiene for woman and baby Postnatal nutrition and physical activity	Assessment of the newborn for danger signs	Exclusive breastfeeding Discuss the importance and management of breastfeeding with women and their families Whole body massage for healthy term infants Early childhood development	When to return for routine contacts Danger signs and care seeking Breastfeeding Warmth for the baby Cord care Sleeping position Washing and bathing Immunization		Non-pharmacological interventions to prevent mastitis Prevention of postpartum constipation Prevention of postpartum anxiety and depression	Clean cord care Sleeping position for the prevention of SIDS	Assessment of newborn danger signs	Midwife-led continuity of care Task shifting components of postnatal care delivery Recruitment and retention of staff in rural and remote areas Involvement of men in PNC and maternal and newborn health Home-based records

Contact	Maternal assessments	Maternal care	Counselling and advice for maternal care	Newborn assessment	Newborn care	Counselling and advice for newborn care	Point-of-care tests	Preventive measures for the woman	Preventive measures for the newborn	Essential good clinical practices	Health system/ health promotion
Contact 3 Day 7–14	Assessment of the woman Screening for postpartum anxiety and depression	Non-pharmacological interventions to treat breast engorgement Postpartum oral iron and folate supplementation Provision of comprehensive contraceptive information and services during PNC	When to return for routine contacts Danger signs and care seeking Lactation Postpartum contraception Sexual activity Prevention of HIV and STIs Prevention of malaria Tobacco and other substance use Postnatal hygiene for woman and baby Postnatal nutrition Physical activity and sedentary behaviour	Assessment of the newborn for danger signs	Exclusive breastfeeding Discuss the importance and management of breastfeeding with women and their families Whole body massage for healthy term infants Early childhood development	When to return for routine contacts Danger signs and care seeking Breastfeeding Warmth for the baby Cord care Sleeping position Washing and bathing Immunization		Non-pharmacological interventions to prevent mastitis Prevention of postpartum constipation Prevention of postpartum anxiety and depression	Sleeping position for the prevention of SIDS	Assessment of newborn danger signs	Midwife-led continuity of care Task shifting components of postnatal care delivery Recruitment and retention of staff in rural and remote areas Involvement of men in PNC and maternal and newborn health Home-based records

Contact	Maternal assessments	Maternal care	Counselling and advice for maternal care	Newborn assessment	Newborn care	Counselling and advice for newborn care	Point-of-care tests	Preventive measures for the woman	Preventive measures for the newborn	Essential good clinical practices	Health system/ health promotion
Contact 4 Week 6	Assessment of the woman Screening for postpartum anxiety and depression	Non-pharmacological interventions to treat breast engorgement Postpartum oral iron and folate supplementation Provision of comprehensive contraceptive information and services during PNC	When to return for routine contacts Danger signs and care seeking Postpartum contraception Sexual activity Prevention of HIV and STIs Prevention of malaria Tobacco and other substance use Postnatal hygiene for woman and baby Postnatal nutrition Physical activity and sedentary behaviour Weight loss	Assessment of the newborn for danger signs	Exclusive breastfeeding Discuss the importance and management of breastfeeding with women and their families Whole body massage for healthy term infants Early childhood development	Breastfeeding Warmth for the baby Cord care Sleeping position Washing and bathing Immunization When to return for routine contacts Danger signs and care seeking		Non-pharmacological interventions to prevent mastitis Prevention of postpartum constipation Prevention of postpartum anxiety and depression	Sleeping position for the prevention of SIDS	Assessment of newborn danger signs	Midwife-led continuity of care Task shifting components of postnatal care delivery Recruitment and retention of staff in rural and remote areas Involvement of men in PNC and maternal and newborn health Home-based record

Contact	Maternal assessments	Maternal care	Counselling and advice for maternal care	Newborn assessment	Newborn care	Counselling and advice for newborn care	Point-of-care tests	Preventive measures for the woman	Preventive measures for the newborn	Essential good clinical practices	Health system/ health promotion
Health worker to provide the service	Doctors, midwives, nurses	Doctors, midwives, nurses	Doctors, midwives, nurses, dieticians, trained community health workers (CHWs)	Doctors, midwives, nurses, dieticians, trained CHWs	Doctors, nurses, midwives, CHWs	Doctors, nurses, midwives	Doctors, nurses, midwives	Doctors, nurses, midwives	Doctors, nurses, midwives, trained CHWs	Doctors, nurses, midwives	Doctors, nurses, midwives, trained CHWs

Annex 4. Antenatal care (ANC) indicators from the monitoring framework

Table A4.1 Core ANC indicators from the monitoring framework

Core indicators				
Monitoring domain	Indicator	Current preferred data source	Other data source(s)	WHO recommendation
Outputs	Percentage of pregnant women with first ANC contact in the first trimester of gestation (before 12 weeks of gestation)	Population-based surveys	Health management information system (HMIS)	Recommendation E.7 ^a
	Percentage of pregnant women who received iron and folic acid supplements for 90+ days	Population-based surveys	HMIS	Recommendations A.2.1 and A2.2 ^a
	Percentage of pregnant women screened for syphilis during ANC	HMIS		Recommendation B.1.7 ^a and Recommendation 4.1 ^b
	ANC contacts: 1. Percentage of women with at least four ANC contacts 2. Percentage of women with at least eight ANC contacts	Population-based surveys		Recommendation E.7 ^a
Outcomes	Percentage of pregnant women who were told about pregnancy danger signs during ANC	Population-based surveys		Recommendation 1 ^c
	Blood pressure measurement: 1. Percentage of pregnant women with at least one blood pressure measure during ANC	Population-based surveys	HMIS	Good clinical practice
	2. Percentage of pregnant women with at least one blood pressure measure in the third trimester during ANC	HMIS		
	Percentage of pregnant women whose baby's heartbeat was listened to at least once during ANC	Population-based surveys		Good clinical practice
	Percentage of pregnant women with an ultrasound scan before 24 weeks	Population-based surveys		Recommendation B.2.4 (updated) ^d
	Experience of care (e.g. waiting time and support received during ANC contacts)	Population-based surveys		Good clinical practice

Adapted from: Lattot SR, Moan AC, Kidula N, Moller A-B, Jayathilaka CA, Diaz T, Tunçalp Ö. Implementation of the new WHO antenatal care model for a positive pregnancy experience: a monitoring framework. *BMJ Global Health*.2020;5(6):e002605 (<https://doi.org/10.1136/bmjgh-2020-002605>). License: CC BY-NC-SA 3.0 IGO.

^a Reference: WHO recommendations on antenatal care for a positive pregnancy experience. Geneva: World Health Organization; 2016 (<https://iris.who.int/handle/10665/250796>).

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^c Reference: WHO recommendations on health promotion interventions for maternal and newborn health 2015. Geneva: World Health Organization, 2015 (<https://iris.who.int/handle/10665/172427>).

^d WHO antenatal care recommendations for a positive pregnancy experience: maternal and fetal assessment update: imaging ultrasound before 24 weeks of pregnancy. Geneva: World Health Organization; 2022 (<https://iris.who.int/handle/10665/352620>). Licence: CC BY-NC-SA 3.0 IGO.

Table A4.2 Context-specific ANC indicators from the monitoring framework

Context-specific indicators				
Monitoring domain	Indicator	Current preferred data source	Other data source(s)	WHO recommendation ^a
Inputs	Health worker density and distribution	Civil registration and vital statistics	Health management information system (HMIS), health-facility surveys and annual administrative reports	Recommendation E.6 – this recommendation applies to rural and remote areas
	Percentage of health units with at least one service provider trained to care for and refer survivors of gender-based violence	Health-facility assessments		Recommendation B.1.3 – this recommendation applies to settings where there is capacity to provide a supportive response (including referral, where appropriate) and that meet the WHO minimum requirements for responding to intimate partner violence and sexual violence against women
Outcomes	Percentage of pregnant women counselled and tested for HIV	Population-based surveys	HMIS	Recommendation B.1.7 – this recommendation applies to high-prevalence settings
	Percentage of pregnant women reporting having received any drug for intestinal worms	Population-based surveys		Recommendation C.4 – this recommendation applies to helminth-endemic areas
	Percentage of pregnant women who receive three or more doses of intermittent preventive treatment for malaria during pregnancy (IPTp 3+)	Population-based surveys	HMIS	Recommendation C.6 – this recommendation applies to malaria-endemic rural areas in Africa
	Percentage of pregnant women who received oral pre-exposure prophylaxis (PrEP)	Population-based surveys	HMIS	Recommendation C.7 – this recommendation applies to pregnant women at substantial risk of HIV infection

^a Reference: WHO recommendations on antenatal care for a positive pregnancy experience. Geneva: World Health Organization; 2016 (<https://iris.who.int/handle/10665/250796>).

Annex 5. Postnatal care (PNC) monitoring indicators

Table A5.1 PNC indicators

Core indicators				
Monitoring domain	Indicator	Current preferred data source	Other data source(s)	WHO recommendation ^a
Outputs	Proportion of women receiving PNC in a facility within a specified time period ^b after delivery ^c	Health facility data	Health management information system (HMIS)	Recommendation C.1
	Proportion of newborns receiving PNC in a facility within a specified time period ^b after delivery ^c	Health facility data	HMIS	Recommendation C.1
Outcomes	Proportion of live births in facilities put to the breast within one hour of birth ^c	Health facility data	Population-based surveys	Recommendation B.5.1 Recommendation B.5.2
	Proportion of women who deliver in a facility and initiate or leave with a modern contraceptive method prior to discharge ^c	Health facility data	Population-based surveys	Recommendation A.6.1
	Proportion of children who have received the first dose of the hepatitis B vaccine in the first 24 hours after birth ^d	Health facility data	Population-based surveys	Recommendation B.2.5

^a Reference: WHO recommendations on maternal and newborn care for a positive postnatal experience. Geneva: World Health Organization; 2022 (<https://iris.who.int/handle/10665/352658>). Licence: CC BY-NC-SA 3.0 IGO.

^b Timing of PNC may vary in accordance with national policy. For births in health facilities, “healthy women newborns should receive postnatal care in the facility for at least 24 hours after birth”. For births at home, the guideline recommends that “the first postnatal contact should be as early as possible within 24 hours of birth”. For healthy women and newborns, the guideline recommends at least three additional PNC contacts: between 48 and 72 hours; between 7 and 14 days; and during week 6 after birth.

^c Reference: Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers. Geneva: World Health Organization; 2023 (<https://iris.who.int/handle/10665/373826>). Licence: CC BY-NC-SA 3.0 IGO.

^d Reference: Hepatitis B first dose vaccine given within 24 hours after birth (%). In: Maternal, newborn, child and adolescent health and ageing: data portal [website]. Geneva: World Health Organization; 2024 ([https://platform.who.int/data/maternal-newborn-child-adolescent-ageing/indicator-explorer-new/mca/hepatitis-b-first-dose-vaccine-given-within-24-hours-after-birth-\(-\)](https://platform.who.int/data/maternal-newborn-child-adolescent-ageing/indicator-explorer-new/mca/hepatitis-b-first-dose-vaccine-given-within-24-hours-after-birth-(-))), accessed 11 October 2024).

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